

Title

Extended Follow-Up and Tolerability Analysis of Photothermal Gold Nanoparticle Therapy for Acne: Laser-Enhanced versus Ultrasound-Assisted Delivery

Abstract

Secondary analysis examined durability of acne improvement and treatment tolerability in the two intervention groups. At extended follow-up assessment, both fractional-laser-primed and ultrasound-enhanced groups maintained clinical improvement gains with minimal recurrence of lesions. Patient-reported satisfaction remained high in both arms. Comparative analysis of treatment-associated discomfort scores revealed similar tolerability profiles. Skin quality assessments using standardised measures indicated durable cosmetic benefit. One participant in each group reported transient post-treatment erythema resolving within expected timeframes. Healing progress was comparable between delivery modalities. Subgroup analysis by baseline severity showed consistent benefit across lesion categories regardless of acne phenotype. No delayed adverse reactions or gold-related systemic effects were observed on repeat laboratory testing. These results support the sustained efficacy of both photothermal treatment strategies for acne, with laser-mediated delivery not conferring additional long-term benefit or tolerability advantage over ultrasound-assisted nanoparticle application.